Capstone Project for PGDAIML
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1. Problem Understanding & Framing

In business, the key problem is understanding the personality and preferences of customer towards a product. Customers have diverse background information, behavior, and perspectives, which are described as ***Features*** contributes to the decision whether to buy a specific product from a company. These features or types of data need to be evaluated in terms of usefulness and purpose.

The problems of companies include ***unknown large customer profiles with complex behavior and product preferences vis a vis limited company budget or resources***. Companies run the business by selling or developing new products blindly to these unknown customers by spending more money, resources and efforts to the entire customers. With these challenges, companies tend to have high product development and marketing costs making the cost to serve high and non-profitable. Company prioritization, therefore, is crucial to the success of business.

Companies face several problems and challenges when selling and marketing their products to ensure higher profitability. First, companies need to understand to ***who they have to sell the product (People).*** Second, companies need to know ***what products to focus more (Products)*** to ensure higher margin or income. Third, companies need to discern ***how to market or promote the product (Promotion)*** like discounting schemes and other promotions such as buy more or bulk promo. And lastly, companies need to decide ***where to sell the product (Placement)*** either in physical stores or online e-commerce such as websites.

The aim of this AI/ML capstone project is to understand the diverse customer profile, personality, and preferences and to provide business insights to ensure better success in targeting or positioning the product or creating a new product and to help the company achieve its financial targets. However, this project will not answer all the business questions but rather provide simple analyses of selected insights from selected features. This may include answering some business questions related to targeting the right customers based on their profile and focusing on appropriate products, promotions, and placement through a customized and most cost-effective marketing strategies and initiatives.

These machine learning analyses will help a company to focus and spend only to the targeted segment that will most likely buy a product. ***The goal is to market the selected product to a particular segment with appropriate and cost-effective promotions and channel approaches*** for higher likelihood of success of the business.

In this kind of problem or dataset about customer personality or purchase behavior, the use of ***Clustering Algorithm in Machine Learning*** is recommended. Unsupervised machine learning model is the type of ML that is primary used for customer personality and segmentation analysis aimed at grouping customers based on behavioral, demographic, and psychographic patterns.

The different customer information and preferences (features) can be analyzed through unsupervised machine learning models like clustering, especially when handling complex and very large datasets to better understand customer preferences. As a result, companies can develop product or focus the marketing initiatives into a financially healthy and sustainable business.

The industry-standard model is K-Means. K-Means Clustering is best suited for standard behavioral & RFM segmentation that is simple to implement, fast, highly interpretable but applicable for spherical clusters only. Clustering often combined with Principal Component Analysis (PCA) for dimensionality reduction and RFM feature engineering, though advanced architectures provide better nuance for complex datasets. Other models like DBSCAN model that is best suited for irregular customer behavior or fraud/anomaly detection but has limitation to varying densities.

The end-to-end workflow includes the following steps.

1. Data Preparation & Preprocessing. The foundation begins with robust feature engineering, transforming raw transactional logs, interaction records, and demographic attributes into structured behavioral metrics such as Recency, Frequency, and Monetary (RFM) values. Once these features are established, comprehensive data preprocessing is applied using scaling techniques like StandardScaler or RobustScaler to normalize numeric disparities (such as income versus recency) alongside one-hot encoding for categorical factors, ensuring distance-based algorithms are not disproportionately skewed by larger numeric scales.
2. Dimensionality Reduction and Cluster Validation. To combat the curse of dimensionality and eliminate multi-collinearity across numerous customer attributes, dimensionality reduction techniques such as Principal Component Analysis (PCA) are applied to extract key latent dimensions while preserving 80–90% of total variance. Building upon this optimized feature space, optimal cluster selection is conducted using mathematical validation notably the Elbow Method and Silhouette Analysis to identify the mathematically optimal and commercially practical number of customer segments. The **success metrics** is evaluated and interpreted based on **Silhouette Score** although other metrics like Accuracy, AUC, or RMSE can be used.
3. Model execution and business profiling. With the verified parameter set, the chosen clustering model is executed to partition customer data points and assign definitive or probabilistic segment labels to every individual profile. Finally, business profiling and persona mapping translate these mathematical clusters into actionable marketing archetypes via cluster mean analyses and demographic summaries, enabling stakeholders to deploy targeted engagement and personalized product strategies. The **business KPIs** can include **cost savings, decrease in marketing cost, and/or increase in sales revenue.**

2. Data Collection & Understanding

The data of Customer Personality Analysis was sourced from Kaggle (Source: https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis).

The Feature types were summarized, which include descriptive statistics, analysis of missing values and outliers as shown in the python codes *(.ipynb file submitted separately*). The data dictionary were also provided. The features and selected relationships were also visualized to better understand the dataset, followed by data cleaning by removing the outliers, specifically from income and meat purchases.

In the given dataset on customer analyses, the features and data dictionary where data dictionary were provided in the *ipynb file*. Features include four main attributes: 1. **Customer profile** with 9 features, 2. **Products expenditure** in the last two years with 6 features, 3. **Promotion** or campaign approaches with 7 features, and 4. **Placement** with 4 features including number of purchases either on the website, catalogue, or physical store.

There are total of **28 data types** including **1 float** (i.e. income), **25 integer or int** (i.e. ID, year of birth, Kidhome, Teenhome, Recency, MntWines, MntFruits, MntMeatProducts, MntFishProducts, NumWebVisitsMonth, AcceptedCmp3, AcceptedCmp4, AcceptedCmp5, AcceptedCmp1, AcceptedCmp2, Complain, Z_CostContact, Z_Revenue, and Response), and **2 objects** (education, marital status).

```
[5 rows x 28 columns]
Dataset Shape: (2240, 28)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2240 entries, 0 to 2239
Data columns (total 28 columns):
 dtypes: float64(1), int64(25), object(2)
 memory usage: 490.1+ KB
```

The features columns that contain text/object data were also identified.

```
Your text columns are: ['Education', 'Marital_Status']

--- Unique values in 'Education' ---
['Graduation' 'PhD' 'Master' 'Basic' 'Cycle']

--- Unique values in 'Marital_Status' ---
['Single' 'Together' 'Married' 'Divorced' 'Widow' 'Alone' 'Absurd' 'YOLO']
```

Customer

1. ID: Customer's unique identifier
2. Year_Birth: Customer's birth year
3. Education: Customer's education level
4. Marital_Status: Customer's marital status
5. *Income: Customer's yearly household income*
6. *Kidhome: Number of children in customer's household*

7. Teenhome: Number of teenagers in customer's household
8. Recency: Number of days since customer's last purchase
9. Complain: 1 if the customer complained in the last 2 years, 0 otherwise

Products

1. *MntWines: Amount spent on wine in last 2 years*
2. *MntFruits: Amount spent on fruits in last 2 years*
3. *MntMeatProducts: Amount spent on meat in last 2 years*
4. *MntFishProducts: Amount spent on fish in last 2 years*
5. MntSweetProducts: Amount spent on sweets in last 2 years
6. MntGoldProds: Amount spent on gold in last 2 years

Promotion

1. NumDealsPurchases: Number of purchases made with a discount
2. *AcceptedCmp1: 1 if customer accepted the offer in the 1st campaign, 0 otherwise*
3. AcceptedCmp2: 1 if customer accepted the offer in the 2nd campaign, 0 otherwise
4. AcceptedCmp3: 1 if customer accepted the offer in the 3rd campaign, 0 otherwise
5. AcceptedCmp4: 1 if customer accepted the offer in the 4th campaign, 0 otherwise
6. AcceptedCmp5: 1 if customer accepted the offer in the 5th campaign, 0 otherwise
7. *Response: 1 if customer accepted the offer in the last campaign, 0 otherwise*

Place

1. NumWebPurchases: Number of purchases made through the company's website
2. *NumCatalogPurchases: Number of purchases made using a catalogue*
3. *NumStorePurchases: Number of purchases made directly in stores*
4. NumWebVisitsMonth: Number of visits to company's website in the last month

3. Data Preprocessing, Applied EDA & Feature Engineering

After understanding the goal and success metrics, collecting and understanding the dataset through **Exploratory Data Analyses (EDA)** in Python was the next critical step for this capstones' machine learning modeling. Exploratory Data Analysis **analyzed, cleaned, and visualized dataset** to uncover patterns, spot anomalies, test hypotheses, and check assumptions. EDA acted like detective work to understand the "story" behind the raw data instead of jumping straight into

machine learning or predictive modeling. **One hot encoding** was used to converts categorical data into a binary numerical format for machine learning models.

Based on the dataset, year of birth and income had potential outliers that were removed in the data. Looking at the visual distribution of the customers, the following findings were observed *(Pls refer to the .ipynb file*).

- Year_Birth: Customer's birth year between 1920 to 2000 and Income: Customer's yearly household income up to 120,000.

Here is a Data dictionary indicating the lower and upper limits that removed outliers.

```
# Data Dictionary and limits
# 1. Specify the feature name and target range
filter_ranges = [
    ['Year_Birth',1920,2000],
    ['Income',0,120000],
    ['MntWines',0, 1000],
    ['MntFruits',0, 150],
    ['MntMeatProducts',0,1000],
    ['MntFishProducts',0,300],
    ['MntSweetProducts',0,300],
    ['MntGoldProds',0,400],
    ['NumDealsPurchases',0,30],
    ['NumWebPurchases',0,30],
    ['NumCatalogPurchases',0,30],
    ['NumStorePurchases',0,30],
    ['NumWebVisitsMonth',0,30]
        ]

# 2. Filter the entire DataFrame based on that specific feature
filtered_df=df.copy()
for feature,lower_limit,upper_limit in filter_ranges:
    filtered_df = filtered_df[filtered_df[feature].between(lower_limit, upper_limit)]

print(filtered_df)
filtered_df.describe()
```

Based on plot distributions, Birth year and most marital status (either single or married or together, or divorce or widow except YOLO, alone & absurd) did not contribute to wine preference. However, high income and non-basic education status showed preference to wines.

In the Promotions plots, higher income tended to accept offer in the first campaign AcceptedCmp1 while Basic education tended not to accept 1st offer AcceptedCmp1.

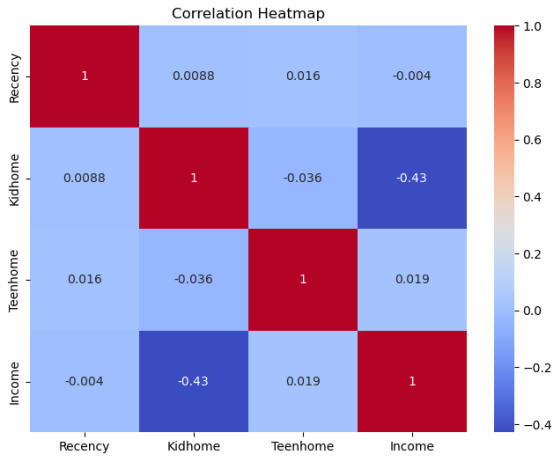
In the Placement plots, Birth year and marital status either single or married (except YOLO, alone, & absurd) did not contribute to NumStorePurchases. However, high income status showed preference to NumStorePurchases. Basic education showed less purchase (NumStorePurchases)

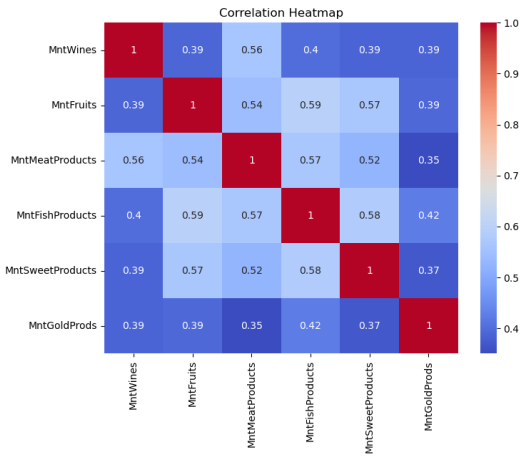
Standard Scaler algorithm was implemented to normalize numerical data so that every feature has a mean of zero and a standard deviation of one. It was used to prevent machine learning algorithms from giving unfair weight to features with larger numeric values (i.e. income as high as 162,397 relative to number of website visit per month with only maximum of 20). It achieves this by shifting all features to a common scale where they share a mean of zero and a variance of one. This uniform scale directly boosts model performance by allowing gradient descent algorithms to converge faster and distance-based models to calculate geometric distances accurately.

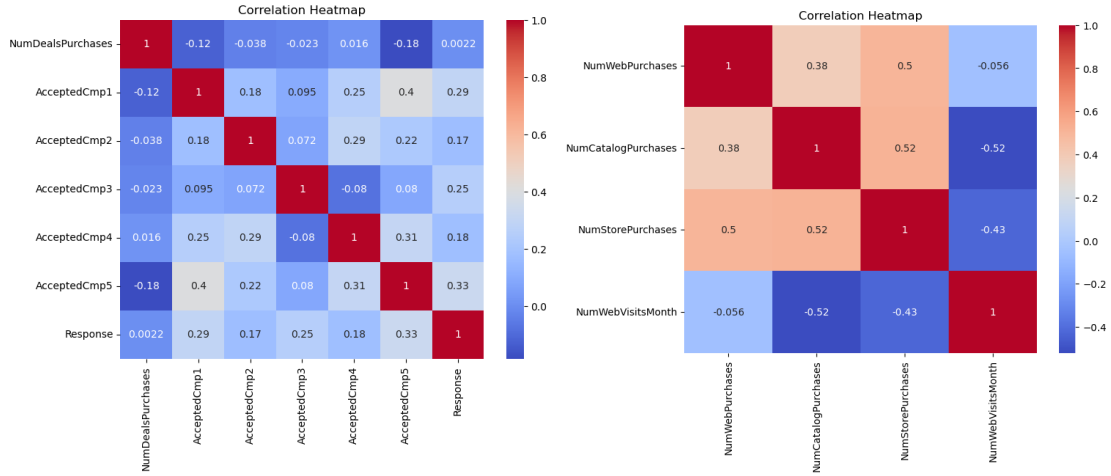
Correlation heatmap

Correlation heatmap graphical tool displayed the strength and direction of relationships between multiple variables in a dataset using color. +1 (Strong Positive Correlation), 0 (No Correlation), and -1 (Strong Negative Correlation). *(.ipynb file submitted separately*).

- On customer profile: Income and Kidhome had high inverse or negative correlation (-0.49). Other features have no or less correlation.
- OnProducts: high positive correlation at 0.59 for MntFruits and MntFishProducts showing healthy food preference. MntFruits and MntSweetProducts at 0.57 and MntWines and MntMeatProducts at 0.56. There is high chance that customers who buys meats also buy wines. Other products also have positive correlation, at least 0.35.
- On Promotion, AcceptedCmp1 (customer accepted the offer in the 1st campaign) and AcceptedCmp5 (customer accepted offer in last campaign) are positively correlated at 0.40
- On Placement, NumCatalogPurchases (number of purchases made using a catalogue) and NumStorePurchases number of purchases made directly in stores) are positively correlated at 0.52. This implies that the catalog helps customers to purchase at the stores. NumCatalogPurchases or number of purchases made using a catalogue is inversely correlated with NumWebVisitsMonth at -0.52. Surprisingly, low correlation between NumWebVisitsMonth and NumWebPurchases (-0.056), meaning number of web site visits does not guarantee web purchase.







Feature selection in Python was used to finding relevant features. There are three frameworks, that are built directly into scikit-learn - Filter methods (statistical metrics), Wrapper methods (iterative testing), and Embedded methods (model-integrated properties). **Embedded** method such as RandomForest or Lasso captures interactions of the features and is optimized for the model unlike Filter method like SelectKBest that is model agnostic, computationally light, but ignores feature interactions.

Under the product features, Wines and Meat Products purchase had similarities while Fruits, **Fish Products**, Sweet Products and Gold Products had similarities.

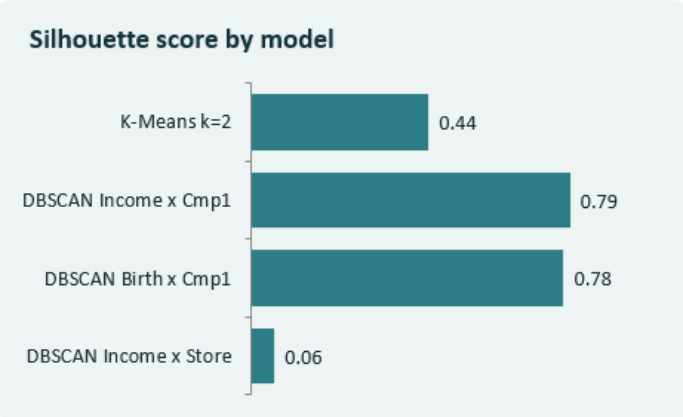
Under the promotion features, the number of deal purchases had a different pattern compared to the other features (AcceptedCmp1, AcceptedCmp2, **AcceptedCmp4**, **AcceptedCmp5** and AcceptedCmp3 and Response) had similar pattern based on visualization.

Lastly under the placement features, NumWebPurchases, NumCatalogPurchases and NumStorePurchases similar increasing trend with higher income while **NumWebVisitsMonth** has inverse relationship with income.

The selected Features were ['feature_10', 'feature_12', 'feature_18', 'feature_22', 'feature_24'].

4. Model Implementation

Unsupervised K-Means and DBSCAN were used and compared. The **Silhouette Score** was used to evaluate the quality of a clustering algorithm by measuring how well-separated and compact the clusters are. It evaluated whether data points have been assigned to the correct groups by giving a single score back for the model. If the data points are well inside their own cluster and far away from neighboring clusters, score range close to 1, can be considered excellent clustering. For example, the features 'Income' and 'AcceptedCmp1' with Silhouette Score (excluding noise) of 0.791 showed good clustering with two clusters. Similarly, 'Year_Birth' and 'AcceptedCmp1' showed good clustering with two clusters that had Silhouette Score (excluding noise) of 0.776. Other features selected like 'Income' and 'NumStorePurchases', the Silhouette Score (excluding noise) is low 0.056.



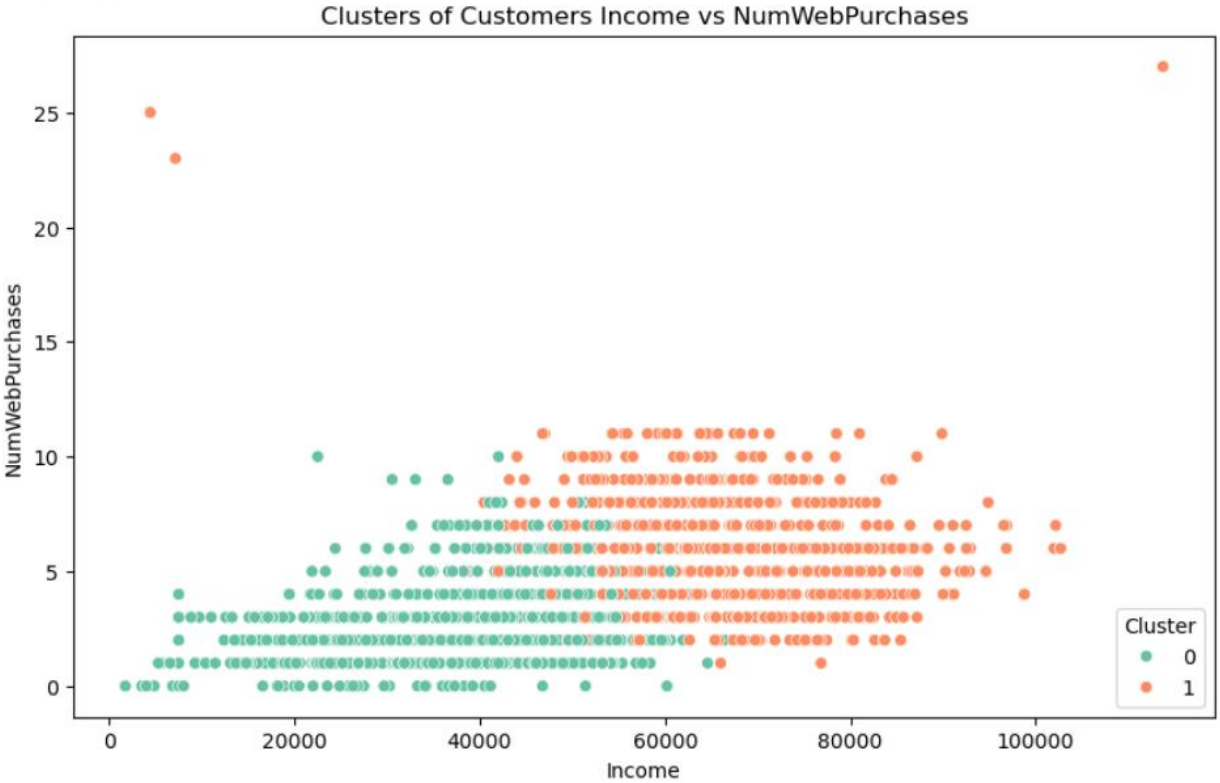

```
Silhouette Score for k=2: 0.44
```



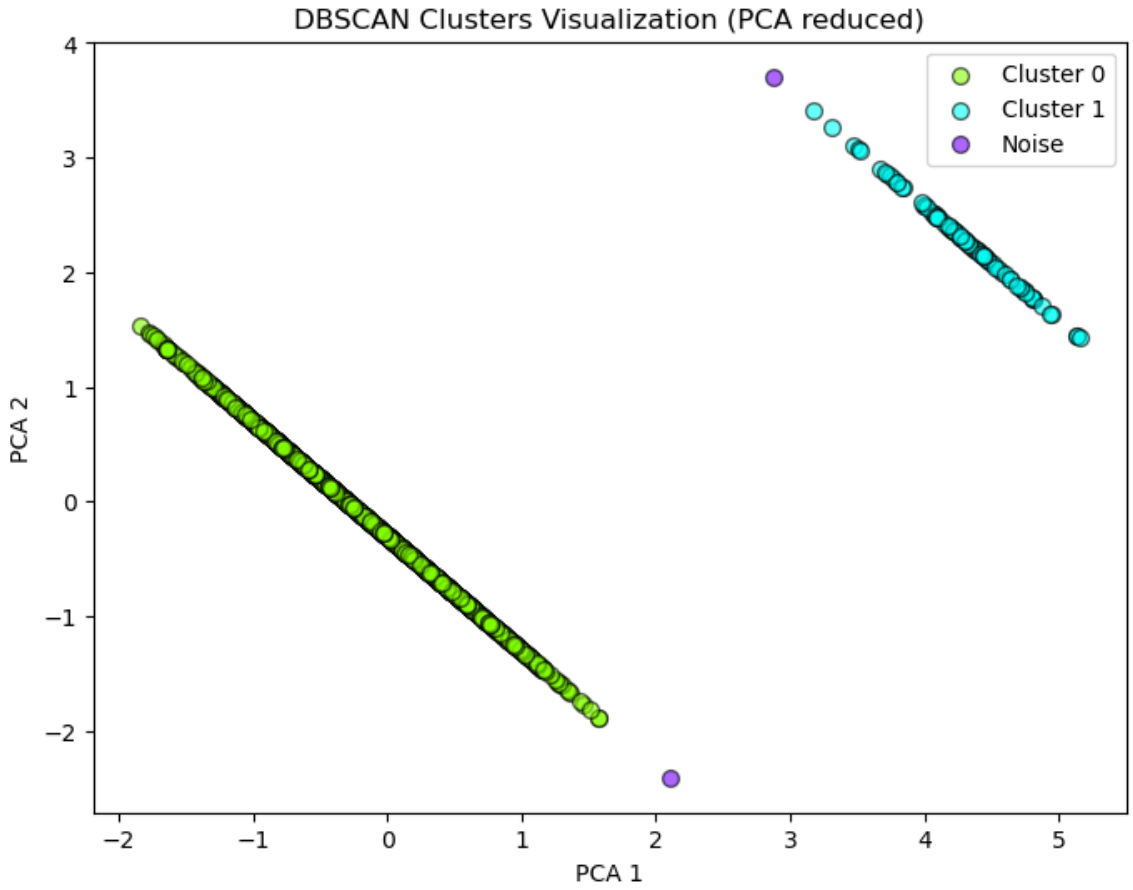
```
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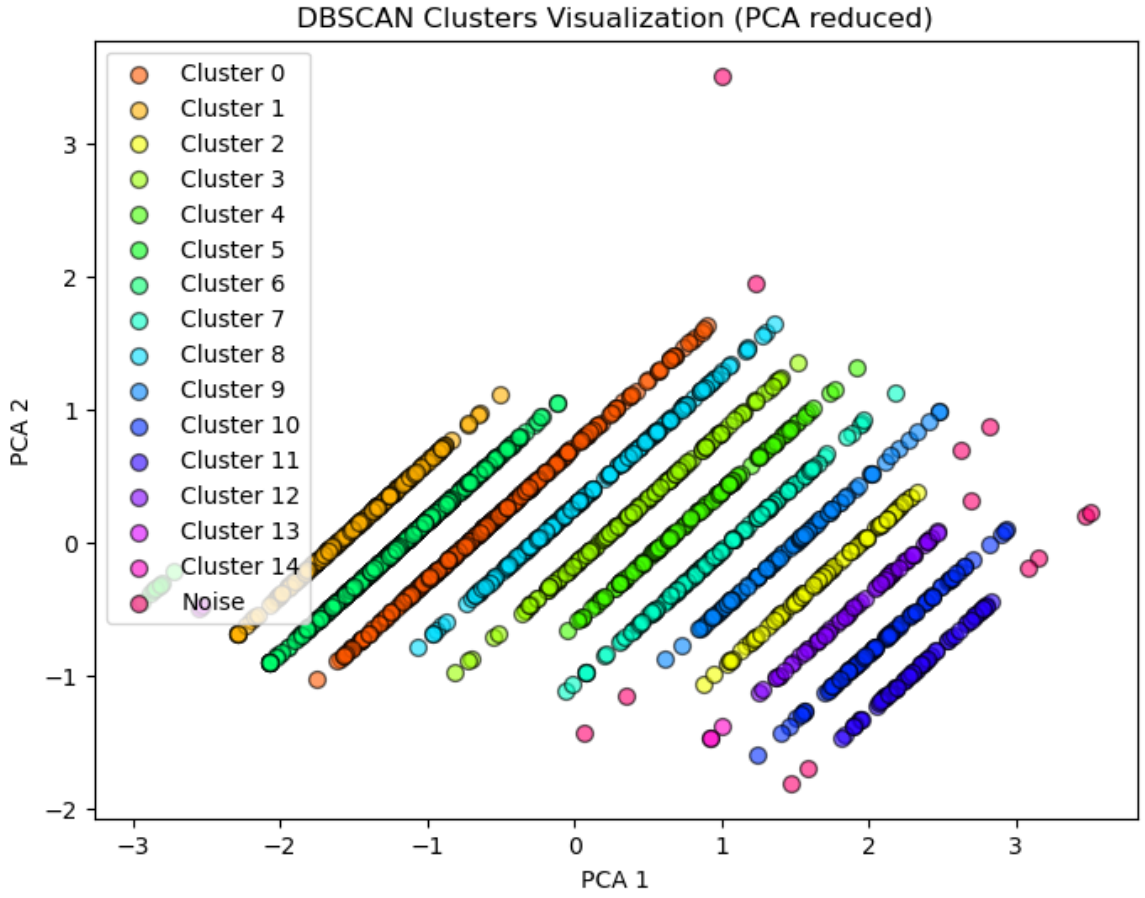


```
Clusters labels:
[-1  0  1]
Silhouette Score (excluding noise): 0.791
```



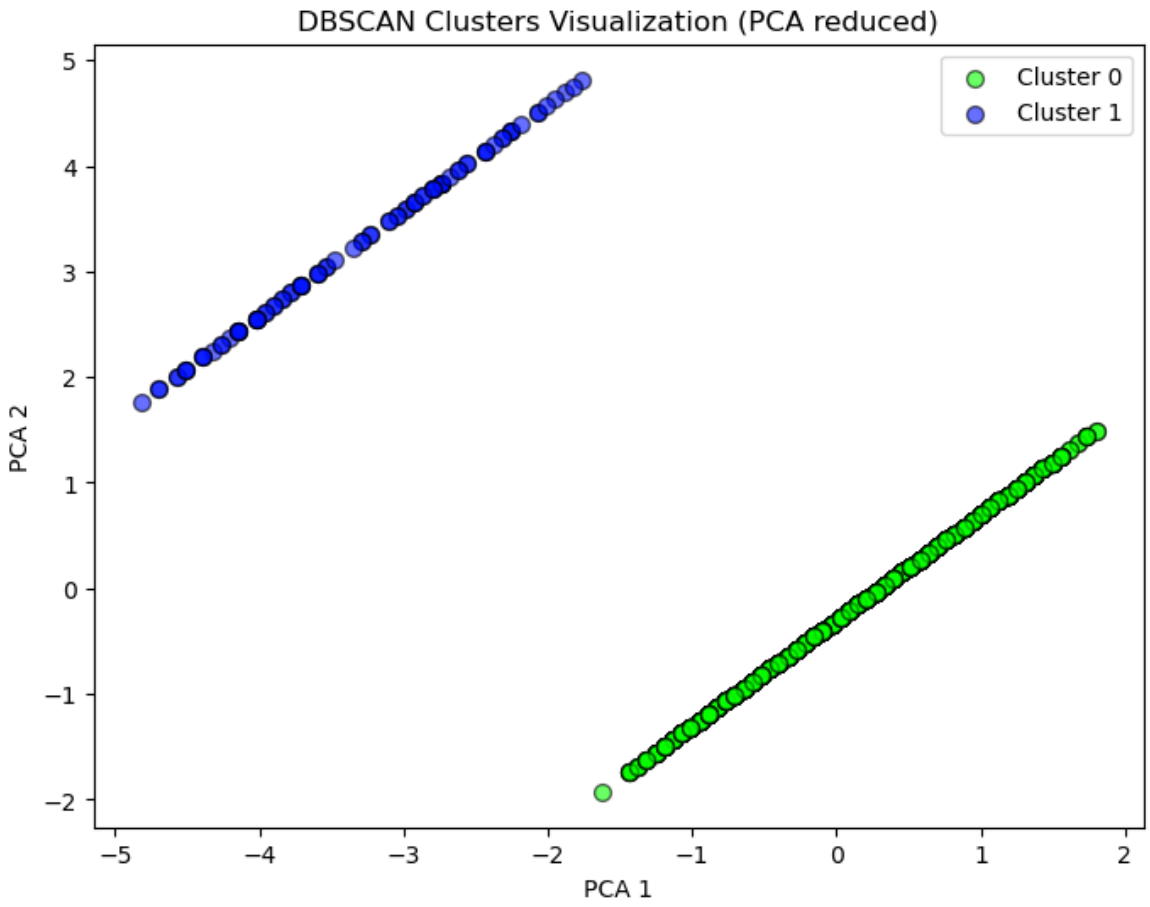

```
Number of clusters found: 2
Number of noise points: 2
                Income  AcceptedCmp1
Cluster
-1       75831.500000           0.5
 0       48377.139188           0.0
 1       78272.280899           1.0
```

```
Clusters labels:
[-1  0  1  2  3  4  5  6  7  8  9 10 11 12 13 14]
Silhouette Score (excluding noise): 0.056
```




```
Number of clusters found: 15
Number of noise points: 13
                Income  NumStorePurchases
Cluster
-1       80821.538462          10.076923
 0       43487.665552           4.000000
 1       30677.440367           2.000000
 2       69521.428571          10.000000
 3       59024.903846           6.000000
 4       63906.484615           7.000000
```

```
Clusters labels:
[0 1]
Silhouette Score (excluding noise): 0.776
```




```
Number of clusters found: 2
Number of noise points: 0
         Year_Birth  AcceptedCmp1
Cluster
0        1969.045175           0.0
1        1968.466667           1.0
```

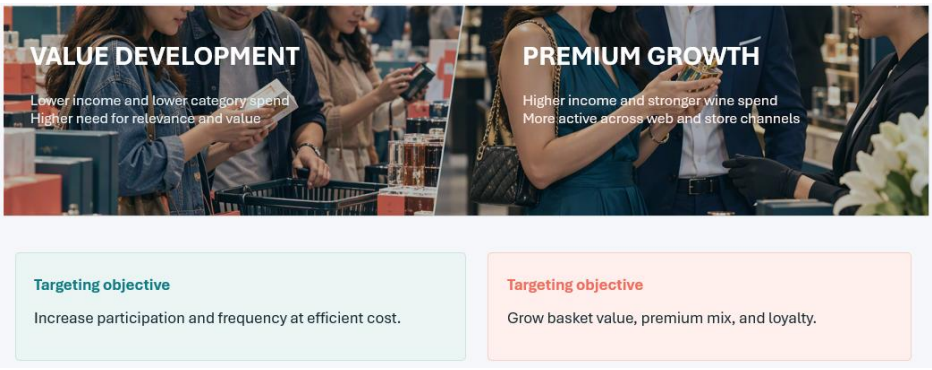
On customer, higher-income customers buy more wine and shop in store more often. Basic-education customers accept fewer offers. On product targeting, wine and meat buyers overlap; fruit, fish and sweets buyers form a second group (0.57-0.59). With this customer segment information, the right product will be focused depending on the said grouping aforementioned.

To maximize sales and reduce cost-to-serve, cross-selling of products is recommended based on the ML insights. Wine and meat buyers overlap. Furthermore, fruit, fish and sweets also move together.

Related to promotions and campaigns, the results showed that customers who accepted the first campaign tend to accept the latest one. And in terms of placement, catalogue and store purchases move together indicating that catalogue helps drives purchase on physical stores. However, website visits do not guarantee web purchases.

Based on the results, customers who accepted the first campaign average $78.3k income versus $48.4k for those who did not. Moreover, a handful of unusual customers (income about $103k, wine spend about $610) deserve individual attention.

Two customer groups require different treatment.



5. Critical Thinking

There are some limitations found which include: 1) Silhouette 0.78-0.79 reflects a binary axis, not natural groups, 2) Imbalance: Response 14.9% positive; AcceptedCmp2 1.3%, and 3) Leakage risk: campaign-response fields as clustering inputs.

The identified bias risks included the following areas – Year Birth, Income and Education, Basic-education customers, Rare labels (YOLO, Absurd, Alone), and no gender or race fields to audit. To responsively use the insights, the potential risks need to be managed. Age, income and education can act as proxies for socioeconomic status. Basic-education customers accepted fewer offers; targeting those risks excluded them. Few customers respond (14.9%), so patterns can be unstable.

To improve the model and insights, there are several steps that are recommended. On the technical side, there is a need to compare segment membership and targeting rate by age band, education, marital status. The metrics can be demographic parity difference, disparate impact (>= 0.8) and explain segments with SHAP on a surrogate classifier. On a business and pragmatic basis, it is recommended to define all business questions and rerun feature selection on all features. Then, pilot a targeted campaign against a control group. Then, track the cost savings, marketing cost and revenue uplift. And finally, review fairness results before scaling.

Leadership guardrails are critical to the implementation and success of customer targeting strategy. Five steps need to be followed. First, an audit is needed. Compare segment membership and targeting rates by age band, education and marital status. Second, need to measure by tracking the demographic parity difference and disparate impact. Third, use a surrogate classifier with SHAP to explain segment assignment drivers. Fourth, experiment first. Pilot targeted campaigns against a control group before scaling. And lastly, monitor the execution and success metrics by reviewing fairness, cost savings, marketing cost and revenue uplift together.